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Widespread inclination shallowing in Permian and Triassic paleomagnetic data from Laurentia: Support from new paleomagnetic data from Middle Permian shallow intrusions in southern Illinois (USA) and virtual geomagnetic pole distributions

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ABSTRACT

Recent paleomagnetic work has highlighted a common and shallow inclination bias in continental redbeds. The Permian and Triassic paleomagnetic records from Laurentia are almost entirely derived from such sedimentary rocks, so a pervasive inclination error will expectedly bias the apparent polar wander path of Laurentia in a significant way. The long-standing discrepancy between the apparent polar wander paths of Laurentia and Gondwana in Permian and Triassic time may be a consequence of such a widespread data-pathology. Here we present new Middle Permian paleomagnetic data from igneous rocks and a contact metamorphosed limestone from cratonic Laurentia. The exclusively reversed Middle Permian magnetization is hosted by low-Ti titanomagnetite and pyrrhotite and yields a paleomagnetic pole at 56.3°S, 302.9°E ($A_{95} = 3.8$, $N = 6$). This pole, which is unaffected by inclination shallowing, suggests that a shallow inclination bias may indeed be present in the Laurentian records. To further consider this hypothesis, we conduct a virtual geomagnetic pole distribution analysis, comparing theoretical expectations of a statistical field model (TK03.GAD) against published data-sets. This exercise provides independent evidence that the Laurentian paleomagnetic data is widely biased, likely because of sedimentary inclination shallowing. We estimate the magnitude of this error from our model results and present and discuss several alternative corrections.

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1. Introduction

It is now well-established that depositional remanent magnetizations (DRMs) of sedimentary rocks are prone to a shallow inclination bias due to the settling of magnetized particles in the gravitational field, which may overcome the vertical torque applied to the particles by the geomagnetic field. When uncorrected, this shallow inclination bias will result in computation of a "far-sided" paleomagnetic pole, or one that is shifted (from the true location) away from the sampling site. The paleogeographic corollary is a low-latitude bias. It has been hypothesized that this bias may be a major contributing factor to the long-standing discrepancy between the Permian and Triassic segments of the apparent polar wander paths (APWPs) of Laurussia and Gondwana, when the landmasses are reconstructed in a conventional way (Fig. 8a; Rochette and Vandamme, 2001; Domeier et al., 2009). Recent paleomagnetic results from Gondwana, derived from volcanic rocks and sedimentary rocks corrected for inclination shallowing, have improved the agreement between these APWPs, thereby supporting this hypothesis (Brandt et al., 2009; Domeier et al., 2011).

Here we consider the possibility that such a shallow inclination bias may also be widely present in the Permian and Triassic paleomagnetic records from Laurentia (as a proxy for Laurussia). After updating the compilation of paleomagnetic results from Torsvik et al. (2008), according to the 2009 geologic timescale (Walker and Geissman, 2009), the Permian–Middle Triassic (300–235 Ma) data-set from Laurentia includes only two poles derived from igneous rocks (of 37 records) (Table 2). The remaining 35 poles are derived from continental redbeds, which may be particularly susceptible to inclination shallowing (Bilardello and Kodama, 2010; Kent and Irving, 2010; Kent and Tauxe, 2005). To check for the presence (and magnitude) of a shallow inclination bias in these Laurentian sedimentary rock-based paleomagnetic records, we compare published data against theoretical virtual geomagnetic pole (VGP) distributions. Additionally, we present new Middle Permian paleomagnetic results from a suite of shallow intrusive rocks — impervious to sedimentary inclination shallowing — from the Illinois Basin, and thus from cratonic Laurentia. Comparison of this paleomagnetic result with ones derived from coeval redbeds offers an independent means to test for a shallow inclination bias in the latter.

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2. Background geology for the paleomagnetic study

Our paleomagnetic study was conducted on a series of intrusive rocks that invade Devonian and Carboniferous sedimentary rocks in the south end of the Illinois Basin, in southern Illinois and western Kentucky (Fig. 1a). The sampling area lies at the junction of the Reelfoot Rift, the Rough Creek Graben, and the Cottage Grove Fault System, and is recognized as one of the most pervasively faulted regions in the mid-continent (Denny et al., 2008). This focused structural activity is likely due to a prominent weakness in the Precambrian basement that developed during a failed episode of rifting in the late Proterozoic–Early Cambrian, which may have nucleated on an even older lower crustal discontinuity (Kolata and Nelson, 1997; Trace and Amos, 1984). In the late Paleozoic a phase of N-NW compression inverted the extensional structures of the Reelfoot Rift and Rough Creek Graben, and resulted in right-lateral strike–slip displacement on the Cottage Grove Fault (Kolata and Nelson, 1997). Magmatism was broadly contemporaneous with this phase of regional shortening, and is represented by dikes that invaded NW-trending tensional fractures, thin sills, diatreme-like breccias generated by the explosive release of magmatic volatiles, and the uplift of Hicks Dome and the Tolu Arch, which were elevated by intrusions in the lower crust (Fig. 1b; Bradbury and Baxter, 1992; Potter et al., 1995; Luczaj, 1998; Fifarek et al., 2001; Denny et al., 2008). Surface exposures, aeromagnetic surveys, and drilling/mining observations have demonstrated that this NW–SE oriented corridor of intrusions is 100+ km long, stretching from the Tabb Fault System in Kentucky to the Cottage Grove Fault System and the Omaha Dome in Illinois (Hildenbrand and Ravat, 1997; McGinnis and Bradbury, 1964; Padgett et al., 2002; Silverman et al., 2003; Sparlin and Lewis, 1994). Although the dikes are recognized across this corridor, perhaps exploiting the structural grain of the Precambrian basement, the breccias are largely restricted to the vicinity of Hicks Dome, and most occur on the dome flanks (Denny et al., 2008).

Owing to extensive surface weathering and low-temperature alteration, compositional classification of the igneous rocks has proven difficult; the alkaline, ultramafic dikes have been variously described as lamprophyres, mica peridotites, and monticellites or alnöites (Baxter and Desborough, 1965; Currier, 1944; Denny et al., 2002, 2008; Koenig, 1956; Sparlin and Lewis, 1994). The dikes are fine-grained to porphyritic and commonly consist of phlogopite, pyroxene, and olivine (pseudomorphed by serpentine), with accessory apatite, titanomagnetite, garnet, perovskite, titanite, and chromite (Baxter and Desborough, 1965; Denny et al., 2002; Fifarek et al., 2001). Secondary carbonate and chlorite are abundant as replacement phases, in addition to serpentine. The poorly exposed and pipe-like breccia bodies at Hicks Dome have been subdivided into “shatter”, “vent”, and “carbonatitic” varieties, according to their intrusive morphology and the texture of matrix and clasts (Bradbury and Baxter, 1992). Shatter breccias are simply fragmented bodies of the host rock, with an absence of far-

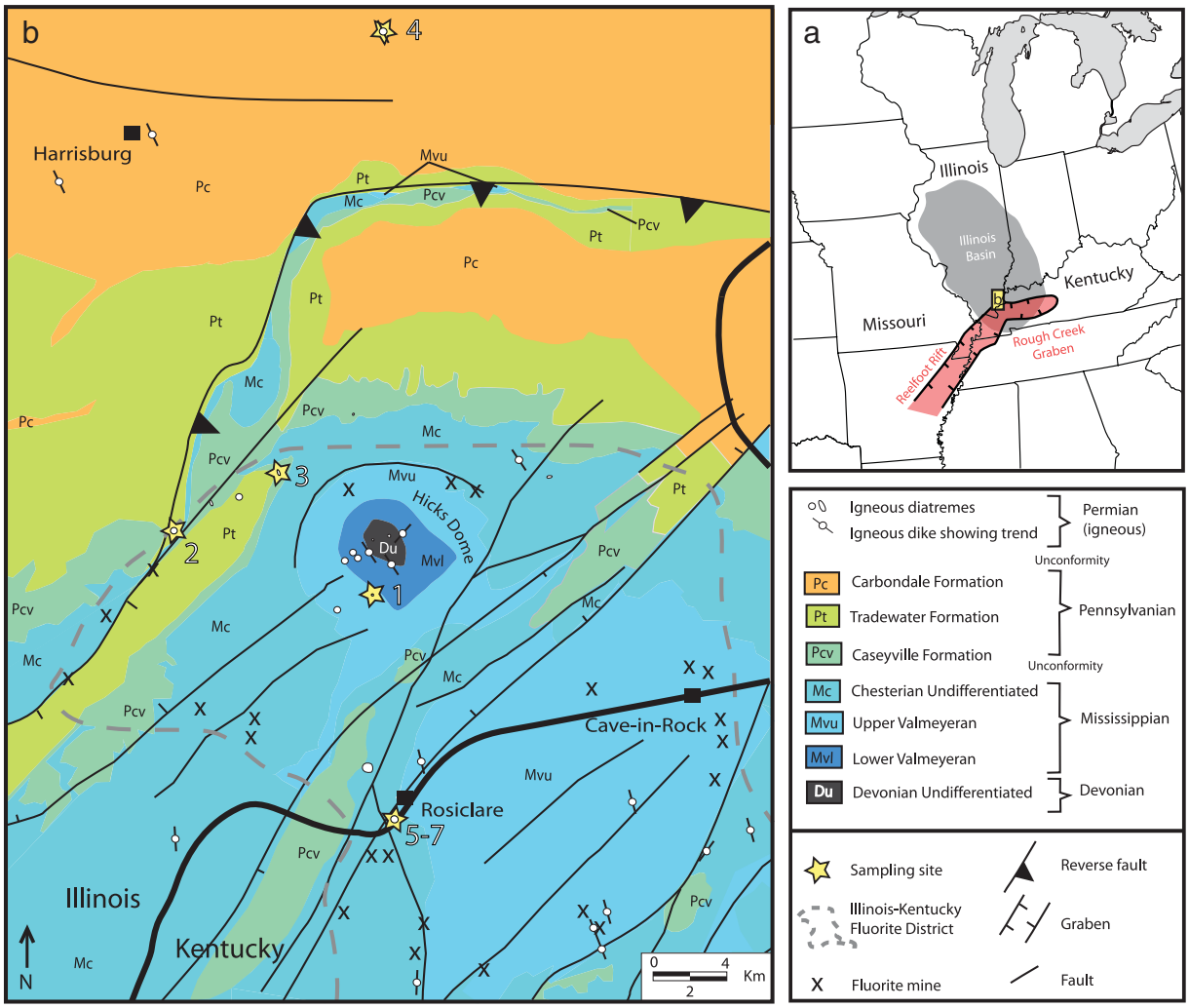


Fig. 1. Geologic map of the southern Illinois sampling area. a) Regional map showing the location of panel b) (highlighted box). b) Geologic map of Hicks Dome and the Illinois–Kentucky Fluorite District. Paleomagnetic sampling sites are labeled with stars and numbered: 1 = Grants Intrusive, 2 = Chamberlain Diatreme, 3 = Hart Creek Diatreme, 4 = Willow Lake Dike, 5 = Downeys Bluff Sill, 6 = Downeys Bluff Limestone (baked), 7 = Downeys Bluff Limestone (unbaked). Simplified from Denny et al. (2008).

traveled or exotic clasts. Vent breccias have xenoliths from underlying sedimentary units, indicating substantial vertical clast-transport. Carbonatitic breccias contain igneous material, in addition to sedimentary clasts, and have a matrix of carbonate. Formation of the latter two breccia varieties is attributed to the explosive expansion of supercritical CO_2-charged fluids as they vaporized during ascent through crustal fractures, after being expelled from alkaline ultramafic magmas at depth (Bradbury and Baxter, 1992; Denny et al., 2008; Luczaj, 1998). The relatively high TiO_2 content and the trace element profile (elevated Zr, Sr, Zn, and REE) of the carbonatitic breccias (Bradbury and Baxter, 1992; Fifarek et al., 2001) is supportive of an alkaline igneous affinity (Erickson and Blade, 1963; Heinrich, 1966).

The age of magmatism has been established as Middle Permian (~270 Ma) on the basis of Rb–Sr, K–Ar, and ^{40}Ar–^{39}Ar geochronology. The following K–Ar and ^{40}Ar–^{39}Ar age estimates have been recalculated with the decay constants of Renne et al. (2010). Rb–Sr results have been recalculated with the decay constants of Steiger and Jäger (1977). Zartman et al. (1967) presented biotite K–Ar dates of 255 ± 13 and 272 ± 13 Ma for a "mica–peridotite" dike and sill, respectively. Rb–Sr dating of the same dike and sill yielded dates of 289 ± 27 Ma and 269 ± 23 Ma. Grants Intrusive, a carbonatitic breccia near Hicks Dome, has yielded K–Ar dates of 261 ± 13 Ma (on biotite) and 284 ± 13 Ma (on hornblende). Reynolds et al. (1997) re-dated the Grants Intrusive by the ^{40}Ar–^{39}Ar method and presented plateau age estimates of 274.5 ± 0.7 Ma (on amphibole) and 275.1 ± 0.7 Ma (on phlogopite). An intrusive breccia in Kentucky produced an average biotite K–Ar date of 276 ± 14 Ma and a Rb–Sr date of 266 ± 22 Ma (Zartman et al., 1967). Fifarek et al. (2001) presented phologopite ^{40}Ar–^{39}Ar isochron age estimates of 271.0 ± 0.3 and 267.8 ± 1.3 from dikes and 269.4 ± 0.4 Ma from an intrusive breccia. The range of mean age determinations, after averaging the results from each intrusion, is 267.8–273.6 Ma. The average mean age of the dikes and intrusive breccias is 270.3 and 271.9 Ma, respectively, suggesting that their emplacement was essentially synchronous.

Subsequent to the late Paleozoic phase of N-NW shortening the regional stress-field again inverted, returning to a tensional regime with extension directed ~NW–SE (Kolata and Nelson, 1997; Trace and Amos, 1984). This regime produced NE-trending, high-angle normal faults, some of which were re-activated Paleozoic structures, that cross-cut and displace the NW-trending dikes and the Tolu Arch. Due to a regional depositional hiatus extending from the Early Permian to the Late Cretaceous, the timing of this extensional episode is poorly resolved (post-Early Permian to pre-Late Cretaceous); however, it has been speculatively attributed to stresses generated by the Late Triassic–Early Jurassic breakup of Pangea (Denny et al., 2008; Kolata and Nelson, 1997). A concentrated fluorite mineralization event is spatially associated with the intrusions and normal faults at the junction of the Reelfoot Rift and the Rough Creek Graben, an area which is also known as the Illinois–Kentucky Fluorite District (Fig. 1b). Fluorite and other ore deposits occur in veins following the NE-trending faults and fractures, and as conformant replacement bodies in Paleozoic limestone (Baxter and Desborough, 1965). The age of the ore bodies and their relationship to the intrusions is debated, but geochemical observations have suggested a genetic connection, as the close spatial association would imply. Specifically, gradients in fluid inclusion temperature and magmatic volatiles (Plumlee et al., 1995; Taylor et al., 1992), and trace and metallic element concentrations (Burruss et al., 1992; Goldhaber et al., 1992) in ore bodies around Hicks Dome suggest that the underlying intrusions acted as thermal and elemental sources for the mineralizing fluids. Attempts to define the age of mineralization have yielded dates of Early Permian (277.0 ± 15.6 Ma by ^{147}Sm/^{144}Nd, Chesley et al., 1994), Early Jurassic (~200 Ma by ^{87}Sr/^{86}Sr, Ruiz et al., 1988), Late Jurassic (~150 Ma by paleomagnetism, Symons, 1994) and Early Cretaceous (~138 Ma by fission track by Harder, 1987, *in* Symons, 1994). If mineralization occurred after the Middle Permian, the temperature of ore precipitation must have been lower than the closure temperature of biotite ($\approx$300°C, Hodges, 1991); otherwise the K–Ar and ^{40}Ar–^{39}Ar ages of the intrusions should have been reset at that time. Fluid inclusion homogenization temperatures in the Fluorite District range from 68 to 227°C in hydrothermal quartz, and 116 to 153°C in fluorite (Richardson and Pinckney, 1984; Spry et al., 1990).

Although there is evidence of more recent movement on some faults in the region — offset Quaternary deposits and recent earthquakes in the neighboring New Madrid Seismic Zone — many faults in the Fluorite District have remained inactive since pre-Late Cretaceous time (Denny et al., 2008; Kolata and Nelson, 1997).

3. Sampling and laboratory methods

Paleomagnetic samples were selectively collected from the freshest available outcrops: a carbonatitic breccia (Grants Intrusive), two vent breccias (Chamberlain Diatreme and Hart Creek Diatreme), an ultramafic dike exposed in an underground coal mine (Willow Lake Dike), and an ultramafic sill (Downeys Bluff Sill, "sill" dated by Zartman et al., 1967), as well as samples of the Downeys Bluff Limestone baked by the sill and others distant from it (Fig. 1b). The paleomagnetism of Grants Intrusive and Downeys Bluff Sill/Limestone has been previously studied by Reynolds et al. (1997).

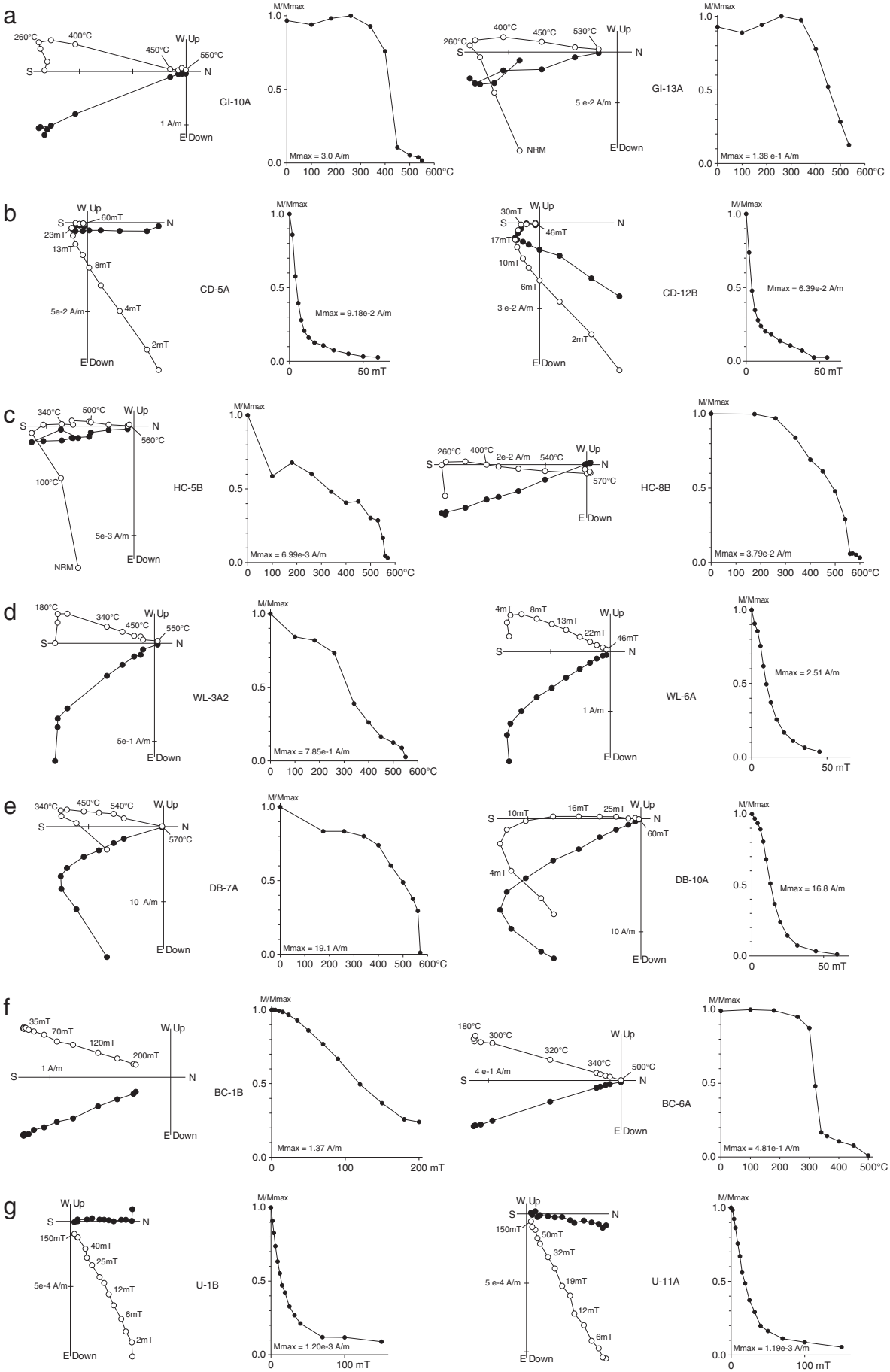
Core samples were collected from the breccias, sill, and limestone with a gasoline powered drill; the orientation of the samples was determined with magnetic and solar compasses. Block samples were collected from the underground dike, from which core specimens were drilled in the laboratory. Sample orientation readings with the magnetic compass were compared with sighted estimates away from the dike, which suggest that local declination anomalies are small ($\leq$3°). Paleomagnetic samples were stored and processed in a magnetically shielded room with a rest field of $\leq$200 nT. Measurements of remanent magnetization were made with a three-axis 2G cryogenic magnetometer. Sister specimens from long cores were subjected to alternating field (AF) and thermal demagnetization techniques in order to determine the most effective demagnetization approach for each site. AF demagnetization was carried out according to a static 3-position procedure. Thermal demagnetization was conducted in air; samples were cooled in a magnetically shielded chamber with a typical DC field of $\leq$5 nT. Demagnetization data were analyzed with orthogonal vector diagrams and stereographic projections (Cogné, 2003; Zijderveld, 1967). Principal component analysis was used to quantitatively define magnetization vectors (Kirschvink, 1980). Fisher (1953) statistics were used to compute site-mean directions from purely vectorial populations; where remagnetization circles defined some samples, the statistical approach of McFadden and McElhinny (1988) was applied.

Hysteresis measurements, conducted on a vibrating sample magnetometer, and a low-temperature remanence experiment, conducted on a magnetic properties measurement system, were performed at the Institute for Rock Magnetism, University of Minnesota.

4. Paleomagnetic results

Both AF and thermal demagnetization of the intrusive breccias revealed the presence of two components of magnetization: a north-directed component angled steeply downward (Component A), and a component inclined horizontally to shallowly upward and directed to the south-southeast (Component B) (Fig. 2a–c). Component A

Fig. 2. Examples of demagnetization behavior. In the orthogonal vector diagrams, the solid (open) symbols are projections onto the horizontal (vertical) plane. All results are presented in geographic coordinates. a) Grants Intrusive, b) Chamberlain Diatreme, c) Hart Creek Diatreme, d) Willow Lake Dike, e) Downeys Bluff Sill, f) Downeys Bluff Limestone (baked contact), g) Downeys Bluff Limestone (unbaked).


a
260°C
400°C
450°C
550°C
W, Up
S
N
1 A/m
E Down
GI-10A
M/Mmax
Mmax = 3.0 A/m
260°C
400°C
450°C
530°C
W, Up
S
N
5 e-2 A/m
NRM
E Down
GI-13A
Mmax = 1.38 e-1 A/m
b
W, Up
60mT
S
N
23mT
13mT
8mT
5e-2 A/m
4mT
2mT
E Down
CD-5A
Mmax = 9.18e-2 A/m
50 mT
30mT
W, Up
S
N
46mT
17mT
10mT
6mT
3 e-2 A/m
2mT
E Down
CD-12B
Mmax = 6.39e-2 A/m
c
340°C
500°C
W, Up
S
N
560°C
100°C
5e-3 A/m
NRM
E Down
HC-5B
Mmax = 6.99e-3 A/m
260°C
400°C
2e-2 A/m
540°C
W, Up
S
N
570°C
E Down
HC-8B
Mmax = 3.79e-2 A/m
d
180°C
340°C
450°C
550°C
W, Up
S
N
5e-1 A/m
E Down
WL-3A2
Mmax = 7.85e-1 A/m
4mT
8mT
13mT
22mT
46mT
W, Up
S
N
1 A/m
E Down
WL-6A
Mmax = 2.51 A/m
50 mT
e
340°C
450°C
540°C
W, Up
S
N
570°C
10 A/m
E Down
DB-7A
Mmax = 19.1 A/m
10mT
16mT
25mT
W, Up
S
N
60mT
4mT
10 A/m
E Down
DB-10A
Mmax = 16.8 A/m
50 mT
f
35mT
70mT
120mT
200mT
W, Up
1 A/m
S
N
E Down
BC-1B
Mmax = 1.37 A/m
200 mT
180°C
300°C
320°C
340°C
500°C
W, Up
4 e-1 A/m
S
N
E Down
BC-6A
Mmax = 4.81e-1 A/m
500°C
g
W, Up
S
N
150mT
40mT
25mT
5e-4 A/m
12mT
6mT
2mT
E Down
U-1B
Mmax = 1.20e-3 A/m
100 mT
W, Up
S
N
150mT
50mT
32mT
5 e-4 A/m
19mT
12mT
6mT
E Down
U-11A
Mmax = 1.19e-3 A/m
100 mT

is the less-stable component, and is most readily removed by AF treatment; it is typically eliminated by ~20 mT or ~400°C. In samples from Grants Intrusive, component B decays to the origin between 400 and 560°C, or by ~80 mT. Two (of 16) samples from this site behaved erratically at high temperature and were excluded from further analysis. Samples from the Chamberlain Diatreme became unstable above 400°C, yielding erratic directions due to thermal alteration, so component B is only defined by AF demagnetization; it is observed to decay by ~60 mT. In samples from the Hart Creek Diatreme, component A is removed more rapidly during the initial demagnetization steps, and holds a greater proportion of the total remanence, relative to the corresponding component from the other intrusive breccias. Component B is largely unblocked between 450 and 560°C, but in some samples a remanence remains above 600°C, and the segment defining component B does not trend to the origin (Fig. 2c). In these samples, a higher laboratory unblocking temperature component (C) can be defined, and it is sub-parallel to component A. This C-component is also observed to persist above 200 mT in some samples that were AF-demagnetized. In two (of 10) samples, only components A and C were identified.

Samples from the Willow Lake Dike and Downeys Bluff Sill yield two components of magnetization that are highly-similar to those identified in the intrusive breccias, and we assign them the same labels (components A and B) (Fig. 2d,e). In the Willow Lake Dike, component A can represent up to 50% of the natural remanent magnetization (NRM) and is removed by ~20 mT or ~300°C, after which component B decays univectorially to the origin. Unblocking of component B is often distributed between 300 and 560°C during thermal demagnetization and is complete by 50 mT during AF demagnetization. Demagnetization of Downeys Bluff Sill samples is complete by 60 mT or 570°C, and cleaning of the lower stability component (A) is complete by ~10 mT or ~300°C. Unblocking of the B-component is typically distributed between 300 and 570°C, but, in a few samples, unblocking at an intermediate temperature (250–450 °C) appears to be distinct from higher temperature decay (500–570 °C), suggesting the presence of a distinct phase. Directional differences between the intermediate and high-temperature vector segments are less than 5°. AF demagnetization spectra do not reveal any correlative distinction in coercivity during removal of the B-component.

Samples of the Downeys Bluff Limestone taken from the baked contact zone adjacent to the ~0.2–0.5 m sill typically preserve only one component of magnetization, sub-parallel to the B-component identified in the sill (Fig. 2f). Unblocking of this magnetization occurs primarily between 260 and 330°C, but a minor fraction of the remanence can persist to 550°C. AF demagnetization is only partly effective; ~30% of the remanence remains after application of a 200 mT peak field. Component A is apparent in a few samples, but it represents a very minor fraction of the NRM and is removed during the initial demagnetization steps. By contrast, unbaked samples of the Downeys Bluff Limestone preserve only the A-component, which decays univectorially to the origin (Fig. 2g). The remanence of the unbaked samples is distinctly less stable, as ~90% of the NRM is removed by application of 100 mT or 300°C.

Considering its relative stability and directional consistency, we assign the B-component the characteristic remanent magnetization (ChRM) of these rocks. We have calculated site-mean directions of this ChRM from 6 of the 7 sites sampled (the unbaked Downeys Bluff Limestone does not exhibit component B) (Fig. 3; Table 1). Within these 6 sites, 87% of the samples (n = 68/78) were retained for the ChRM site-mean calculations. 13% of the samples were rejected due to thermal alteration, erratic behavior, or anomalous directions. The combined site means give the following paleomagnetic pole: 54.1°S, 300.7°E $A_{95} = 4.8°$, N = 6. If tilt-corrections are applied, the pole (hereafter "IL") is located at 56.3°S, 302.9°E ($A_{95} = 3.8°$). Our results are in excellent agreement with the earlier findings of Reynolds et al. (1997); site-mean directions common to both studies (from Grants Intrusive and Downeys Bluff Sill/Limestone) differ by ≤2.2°.

5. Magnetic mineralogy

Hysteresis measurements reveal that the Chamberlain Diatreme, Grants Intrusive, Downeys Bluff Sill, and the Willow Lake Dike are dominated by a low-coercivity magnetic phase (Fig. 4a,b). In conjunction with the thermal demagnetization observations (wherein the ChRM was typically removed by 560°–570°C), and known mineralogy of these igneous bodies, we interpret the ChRM of these rocks to be carried by low-Ti titanomagnetite. Reynolds et al. (1997) reported evidence of low-Ti titanomagnetite in Grants Intrusive and Downeys Bluff Sill from thermomagnetic analysis. There are no strong indications of a second magnetic component in the hysteresis behavior of samples of these rocks, suggesting that the low-temperature/coercivity A-component is held by a less-stable sub-population of titanomagnetite, perhaps distinct in composition or size. Specimens from the contact metamorphosed Downeys Bluff Limestone show both low- and high-coercivity components in variable proportions, indicating that some magnetic phases may be heterogeneously distributed relative to the chip-size used in the experiments. The low-coercivity phase is again consistent with low-Ti titanomagnetite, according to the thermal demagnetization observations. The high-coercivity phase may be the principal ChRM carrier, observed to unblock between 260 and 330°C. Reynolds et al. (1997) interpreted this phase as pyrrhotite, which has a high-coercivity and a Curie point of ~325°C (Rochette et al., 1990). The hysteresis behavior of a specimen from the Hart Creek Diatreme is controlled by a high-coercivity phase; demagnetization of a sister-specimen (from the same sample) shows the NRM of this sample is dominated by the C-component. The persistence of the C-component above 600°C and its association with the observed high-coercivity behavior suggests it may be carried by hematite. The lower temperature components (A and B) are again attributed to titanomagnetite, although the low-coercivity phase in the measured specimen — evident by the "goose-necked" shape of the loop (Tauxe et al., 1996) — is minor.

Low-temperature remanence data from a specimen of the baked Downeys Bluff Limestone supports the interpretation that its remanence is carried by pyrrhotite. The specimen was imparted with an isothermal remanent magnetization at room-temperature and cycled down to 10 K and back, during which time regular measurements of remanence were made. Notable changes in remanence occurred at ~120 K and ~32 K, which correspond to known magnetic transition temperatures in magnetite and pyrrhotite, respectively (Fig. 4c).

6. VGP distribution analysis

It is commonly assumed that the distribution of VGPs is approximately circularly symmetric at all latitudes (but with increasing scatter pole-ward, due to the relative intensity of non-dipole fields) (Harrison, 2009; Tauxe and Kent, 2004). Due to the non-linear relationship between VGPs and magnetic field directions, the corollary of this latitude-independent circular symmetry of VGPs is the expectation of non-circular and latitude-dependent distributions of directions (except at high-latitudes) (Creer et al., 1959; Tanaka, 1999). Tauxe and Kent (2004) used a statistical model of the field to define this theoretical distribution of directions as a function of latitude and insightfully proposed that it could be used to identify departures from the expected distribution of a set of directions generated by a geocentric axial dipole (GAD) at any given latitude. In practice, this is achieved by comparing the observed "elongation" of a directional data-set with the expected elongation from the theoretical function at a specific latitude defined by the mean inclination of the data-set. Elongation is defined by Tauxe (1998) as the ratio of the intermediate and minimum eigenvalues (τ_2/τ_3) of the orientation matrix of the magnetization directions. This so-called elongation–inclination (E/I) technique has led to an increasing recognition of inclination shallowing in sedimentary rocks (Kent and Tauxe, 2005; Krijgsman and Tauxe, 2006; Yan et al., 2005) and to discussions regarding the

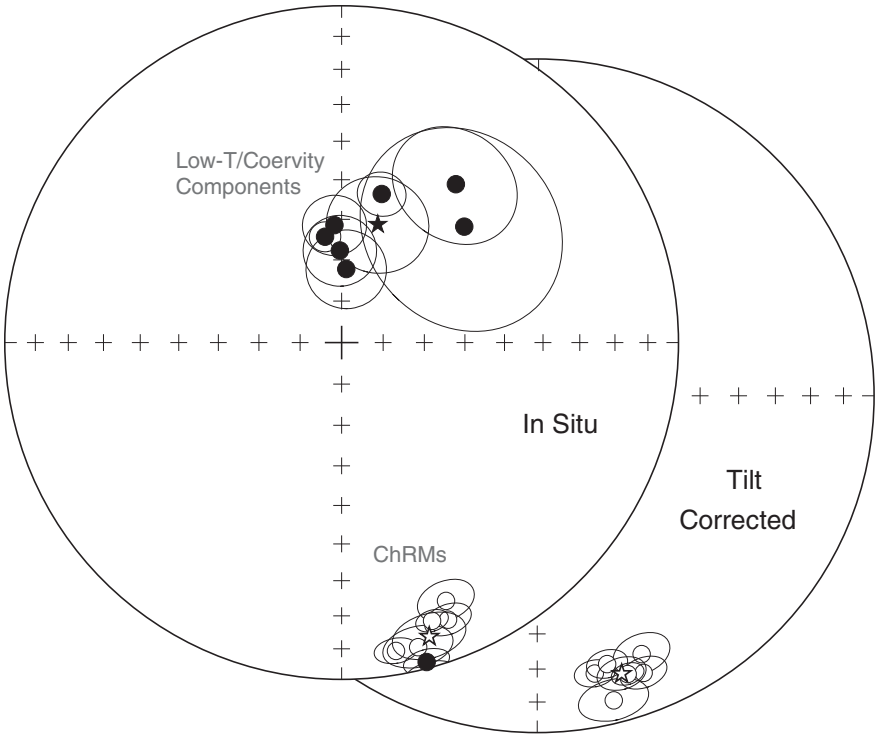


Fig. 3. Directions of site-mean ChRMs (component B) and low-temperature/coercivity magnetizations (component A) and their projected α_{95}. Solid (open) symbols are projections onto the lower (upper) hemisphere. The stars represent the mean of the site means.

structure of the geomagnetic field through time (Tauxe, 2005; Tauxe and Kodama, 2009).

The principal drawback of the E/I technique is that a large amount of data (≥100–150 site-level directions) are required to conduct a meaningful analysis, because the method assumes that the data-set fully captures the directional variance due to secular variation (Tauxe et al., 2008). Small data-sets, therefore, may exhibit apparent elongations reflecting an under-sampling of the short-term field variance, rather than a true departure from GAD expectations. Unfortunately, paleomagnetic studies with 100+ independent site-mean directions are currently rare; there are none within the paleomagnetic records we are considering here (Table 2). Thus, none of the individual studies present a sufficiently large data-set for a rigorous directional distribution analysis. Yet, collectively, these studies constitute an ample data-set that likely represents a full characterization of secular variation. Because several of these studies were conducted on the same formations (e.g. Moenkopi Fm., Chugwater Fm., Cutler Fm.), and many more are derived from lithologically similar continental redbeds, we postulate that it is reasonable, although admittedly not ideal, to combine and collectively analyze these data-sets for common/widespread departures from GAD expectations, as they may be similarly biased. The latitude-dependence of the expected distribution of magnetization directions prevents us from combining data-sets with

Table 1
Paleomagnetic results. N/n/v = number of samples measured/number of samples used in site-mean calculation/number of results defined by a vector (n–v = number of results defined by great-circles). Dg/Ig = declination/inclination in geographic coordinates. Ds/Is = declination/inclination in tilt-corrected coordinates. k = precision parameter of Fisher (1953). α_{95} = half-angle (radius) of the cone of 95% confidence about the site-mean direction. Slat/Slon = latitude/longitude of sampling site. Plat/Plon = latitude and longitude of the site-level VGP. A_{95} = half-angle (radius) of the cone of 95% confidence about the site-level VGP.

Site	N/n/v	Dg	Ig	Ds	Is	k	α_{95}	Slat (°N)	Slon (°W)	Plat	Plon	A95
ChRM directions												
GI	16/14/9	170.0	−7.8	168.3	−16.9	95.8	4.0	37° 30.49	88° 22.59	−59.3	294.8	4.1
WL	9/8/8	158.0	−18.4	158.0	−18.4	90.9	5.8	37° 49.10	88° 20.80	−55.4	312.3	6.1
HC	10/8/6	165.9	−7.5	165.9	−7.5	77.1	6.5	37° 33.92	88° 26.44	−53.8	295.8	6.5
CD	16/14/4	165.1	2.1	165.4	−18.8	100.5	4.1	37° 32.76	88° 28.54	−59.2	300.5	4.3
DB sill	14/13/7	159.0	−12.9	159.0	−12.9	87.4	4.6	37° 24.82	88° 20.85	−53.7	308.6	4.6
DB-baked	13/11/11	162.0	−14.2	162.0	−14.2	212.2	3.2	37° 24.82	88° 20.85	−55.7	304.6	3.2
Mean	6	163.4	−9.8	163.1	−14.8	193.4	4.8	Uncorrected:		−54.1	300.7	4.8
								Tilt corrected:		*−56.3*	*302.9*	*3.8*
Low temperature/coercivity components												
GI	16/11/11	3.6	72.2	327.2	81.2	23.7	9.6	37° 30.49	88° 22.59	70.1	277.4	16.9
WL	9/7/7	46.6	48.5	46.6	48.5	7.0	24.5	37° 49.10	88° 20.80	50.8	359.6	32.2
HC	10/7/7	358.9	67.7	358.9	67.7	49.1	8.7	37° 33.92	88° 26.44	76.9	268.5	14.5
CD	16/14/14	15.1	52.3	42.6	66.3	52.2	5.6	37° 32.76	88° 28.54	76.8	17.9	7.6
DB sill	14/12/12	35.8	41.6	35.8	41.6	9.2	14.5	37° 24.82	88° 20.85	47.9	52.8	17.7
DB-baked	13/7/7	356.5	61.4	356.5	61.4	66.5	7.5	37° 24.82	88° 20.85	84.2	245.3	11.5
DB-unbaked	12/12/12	351.1	64.0	351.1	64.0	141.8	3.7	37° 24.82	88° 20.85	79.4	235.9	5.8
Mean	7	17.0	59.9	19.9	63.9	26.2	12.0	*Uncorrected:*		*79.7*	*349.7*	*18.4*
								Tilt Corrected:		76.5	325.0	22.5

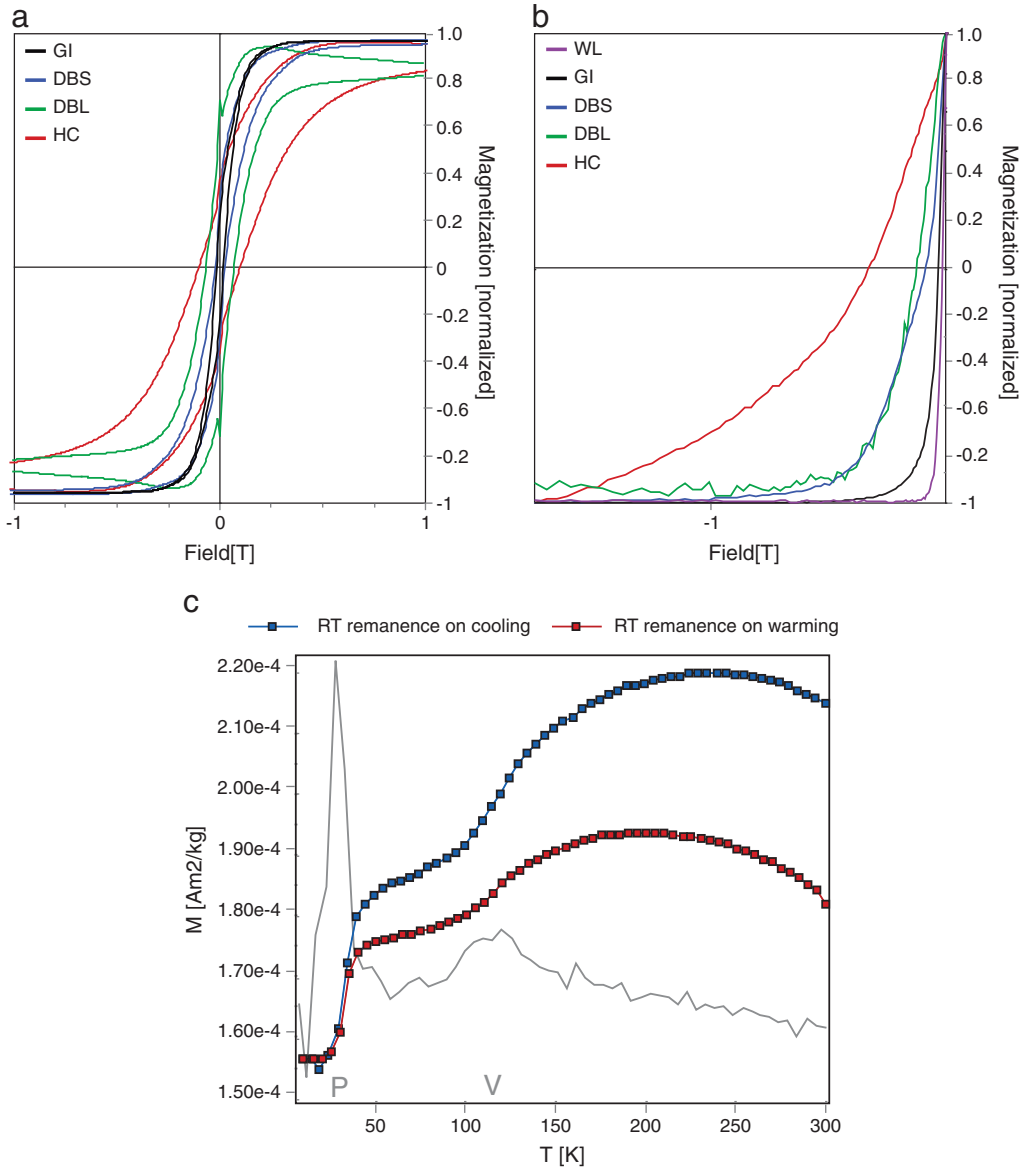


Fig. 4. Results of rock magnetic experiments. Hysteresis loops, after paramagnetic correction a), and back field curves b) of representative samples. Loops were driven to 1.5 T, but the figure is truncated to highlight the low-coercivity behavior expressed by most samples. WL = Willow Lake Dike, GI = Grants Intrusive, DBS = Downeys Bluff Sill, DBL = Downeys Bluff Limestone (baked zone), HC = Hart Creek Diatreme. Results from the Chamberlain Diatreme are not depicted, but are similar to those of Grants Intrusive. Note the "goose-necked" behavior of the HC loop, which may be due to the mixing of magnetite (low-coercivity) and hematite (high-coercivity) signals (Tauxe et al., 1996). c) Low-temperature remanence experiment on a specimen from the baked zone of Downeys Bluff Limestone. Blue and red curves show the change in remanence during cooling and warming, respectively. The gray curve is the first-derivative of the cooling curve; interpreted transitions are labeled: P (pyrrhotite) and V (Verwey).

distinct mean inclinations, thereby significantly limiting our analysis. For this reason, we instead elect to analyze VGP distributions, which have a latitude-independent (consistently circular) expected distribution. As aforementioned, there is an increasing element of VGP scatter with increasing latitude, but we argue that this effect is negligible given that the data come from a narrow range of low-latitude sites.

Because we are specifically interested in testing the hypothesis that the Laurentian data-sets have been systemically biased by inclination shallowing, we employ the statistical field model TK03 (Tauxe and Kent, 2004) to simulate the effect of shallowing on VGP distributions expected from an ideal GAD field. As such, we assign the non-dipole terms in the statistical model a mean of 0 (i.e. TK03.GAD). Shallowing of the ideal synthetic directions follows the relationship determined by King (1955): $\tan(I_o) = f \tan(I_f)$, where f is the "flattening" coefficient, and I_o and I_f are the observed and true field inclinations, respectively. Laboratory re-deposition experiments, E/I analyses, and anisotropy measurements of hematite-bearing rocks have found a mean f of ~0.55, and values as low as $f = 0.4$ ($f = 1$ results in no shallowing ($I_o = I_f$)) (Bilardello and Kodama, 2010; Kent and Tauxe, 2005; Tauxe and Kent, 1984). At 5° intervals between latitudes 0° and 85° we have: (1) drawn 300 directions from TK03.GAD with a mean declination of 0° (no reversals), (2) shallowed the directions by f, (3) transformed the directions to VGPs, and (4) calculated the mean VGP latitude, elongation of the data-set, and the direction of the elongation. This process was repeated 1000 times at each 5° interval to determine the variance in the calculated parameters. The complete experiment was run under three different shallowing conditions: $f = 1$, 0.55, and 0.4; the results are presented in Fig. 5.

To compare the Laurentian paleomagnetic records with the model results, we rotated selected data-sets into the same reference frame according to the following steps: (1) a mean direction (principal eigenvector) is calculated from the orientation matrix (Tauxe, 1998) of the published site-level mean directions, (2) the mean directions are rotated to a declination of 0° (reversed polarity magnetization directions are first inverted) around a vertical axis, (3) the directions are transformed to

Table 2
Compilation of Laurentian paleopoles used in APWP construction. Plat/Plon = latitude and longitude of published paleopole. $Plat_{0.54}$/$Plon_{0.54}$ = latitude and longitude of paleopole after inclination correction using $f = 0.54$. $Plat_{0.46}$/$Plon_{0.46}$ = latitude and longitude of paleopole after inclination correction using $f = 0.46$. A_{95} = half-angle (radius) of the cone of 95% confidence about the paleopole. Age = best estimate of age of magnetization (updated from Torsvik et al., 2008). Notes: D = evidence presented in original reference suggests that the characteristic remanent magnetization (ChRM) may be a depositional remanent magnetization (DRM). C = ChRM is explicitly interpreted by original authors to be a chemical remanent magnetization (CRM). ? = insufficient evidence is presented in the original reference to draw a conclusion about the nature of the ChRM. Reference = global paleomagnetic database reference lookup number; M = Molina Garza et al. (2000), W = Wynne et al. (1992).

#	Rock unit	Plat	Plon	$Plat_{0.54}$	$Plon_{0.54}$	$Plat_{0.46}$	$Plon_{0.46}$	A95	Age	Notes	Reference
Sedimentary records											
1	Upper Red Peak Fm.	−49.0	285.0	−53.5	288.5	−55.2	290.0	5.0	235	D	1134
2	Lower Fundy Group, Nova Scotia[a]	−45.0	277.0	−46.3	276.6	−46.8	276.4	7.2	235	D	2266
3	Moenkopi Fm., Anton Chico Mem.[a]	−45.0	301.0	−45.6	301.6	−45.9	301.8	4.9	243	D	2979
4	Moenkopi Fm., Anton Chico Mem.[a]	−40.0	307.0	−39.5	306.4	−39.4	306.2	5.3	243	D	2632
5	Moenkopi Fm.	−56.0	285.0	−60.3	289.7	−61.8	291.8	4.5	243	D	2808
6	Moenkopi Fm.[a]	−56.0	289.0	−60.0	293.6	−61.4	295.5	3.4	243	D	2489
7	Chugwater Fm.[a]	−45.0	295.0	−49.8	299.5	−51.5	301.4	4.0	245	D	1266
8	Chugwater Fm.[a]	−47.0	294.0	−52.3	299.1	−54.2	301.3	3.3	245	D	1271
9	Moenkopi Fm.	−53.0	291.0	−58.5	297.2	−60.4	300.0	3.1	245	?	159
10	Moenkopi Fm. (Gray Mtn.)	−55.0	286.0	−59.5	291.3	−61.2	293.6	5.0	245	D	1221
11	Lower Red Peak Fm.	−46.0	301.0	−51.7	307.6	−53.7	310.3	7.0	245	D	1134
12	Moenkopi Fm. (drill core)	−55.0	289.0	−61.3	296.5	−63.4	300.0	2.5	247	?	160
13	Moenkopi Fm.	−56.0	279.0	−61.7	283.8	−63.7	286.1	4.9	247	C	571
14	Dewey Lake Fm. (Texas)[a]	−51.0	306.0	−54.5	310.7	−55.8	312.7	5.0	252	D	2303
15	Dewey Lake Fm. (New Mexico)[a,b]	−55.0	298.0	−58.2	302.1	−59.3	303.9	6.0	252	D	M
16	Bernal Fm.	−50.0	300.0	−52.5	303.2	−53.4	304.5	8.0	255	D	2489
17	Ochoan redbeds	−55.0	299.0	−59.5	305.3	−61.1	308.1	15.0	255	?	688
18	Upper Maroon Fm.[a]	−58.0	292.0	−66.2	304.2	−68.8	310.4	12.8	256	?	504
19	Guadalupian redbeds	−51.0	305.0	−54.7	310.2	−56.0	312.4	5.0	265	?	688
20	Toroweap Fm.	−52.0	305.0	−56.1	311.2	−57.5	313.7	10.0	273	?	688
21	Leonardian subset	−52.0	299.0	−55.0	302.6	−56.0	304.1	5.0	275	?	688
22	Artinskian Pictou redbeds[a]	−42.0	306.0	−41.0	305.9	−40.6	305.8	3.6	280	?	2281
23	Fountain and Lykins Fms.[a]	−45.0	306.0	−49.4	311.2	−50.9	313.4	13.1	285	?	504
24	Abo Fm.	−47.0	305.0	−49.3	308.1	−50.2	309.3	2.1	286	D	1311
25	Cutler Fm.[a]	−41.0	302.0	−42.5	303.6	−43.1	304.3	2.0	287	C	671
26	Cutler Fm.	−40.0	308.0	−43.0	311.7	−44.1	313.2	12.3	287	?	675
27	Cutler Fm. (Lison Valley)	−41.0	308.0	−44.0	311.7	−45.1	313.2	7.1	287	C	1341
28	Upper Casper Fm.	−51.0	303.0	−58.3	313.1	−60.6	317.7	1.5	289	C	1455
29	Elephant Canyon Fm	−42.0	302.0	−44.1	304.3	−44.8	305.2	5.0	289	C	671
30	Ingelside Fm.	−43.0	308.0	−47.2	312.9	−48.7	314.8	2.0	292	C	1142
31	Minturn and Maroon Fms.	−40.0	301.0	−40.5	301.4	−40.6	301.5	2.8	298	?	1685
32	Dunkard Fm.	−44.0	303.0	−41.0	301.7	−39.8	301.2	3.9	300	?	302
33	Laborcita Fm.	−42.0	312.0	−43.2	313.5	−43.6	314.1	2.1	300	D	1311
34	Prince Edward Is. redbeds	−42.0	313.0	−41.7	312.9	−41.6	312.9	6.0	300	?	336
35	Prince Edward Is. redbeds	−41.0	306.0	−39.2	305.8	−38.5	305.7	5.8	300	?	276
Igneous records											
36	Cross Diatreme, British Columbia[b]	−49.0	295.0	–	–	–	–	6.0	245	–	W
37	Basic Sill, Prince Edward Is.	−52.0	293.0	–	–	–	–	5.0	252	–	431
38	Southern Illinois Intrusives ("IL")[b]	−56.3	302.9	–	–	–	–	3.8	270	–	This study

[a] Result included in the VGP distribution analysis.
[b] Result not listed in Torsvik et al. (2008).

VGPs according to the dipole formula, (4) the sampling sites (and thus the VGPs) are rotated to a longitude of 0° around a vertical axis, (5) a mean VGP is calculated from each VGP set (again using principal component analysis), and (6) each VGP set is rotated around a horizontal axis (trending E–W) until the mean VGP is vertical. The results of this exercise are shown in Fig. 6a. Also depicted in Fig. 6 are the elongations and their declinations for each record. We stress that the small number of site-level VGPs in each of these individual records ($N \leq 36$) means that these elongation values are not likely representative of the "true" elongation that would be derived from a data-set that fully captured the directional variance of a time-averaged field. Again, it is for this reason that we combine and collectively analyze these individual records, according to the argument that the lithologically similar redbeds may have been similarly affected by inclination shallowing. To minimize the bias introduced by the centering process, we exclude data-sets with <8 listed site-level directions, which reduces our compilation to 14 records with a total of 285 site means.

The magnitude and direction of the elongation of the combined data-set were calculated (Fig. 7b); confidence bounds on the estimate of the elongation were determined via 1000 re-calculations of elongation from randomly sampled subsets of the original data (i.e. bootstrap). A fundamental assumption in this analysis is that all VGPs belong to the same general population, meaning that the presence of secondary magnetizations or incompletely removed overprints may significantly bias the elongation calculation. For this reason, we repeated the analysis after excluding three records from the study of McMahon and Strangway (1968), where AF demagnetization was used sparingly, and no samples were subjected to thermal or chemical demagnetization (Fig. 7c). Thermal and/or chemical demagnetization is necessary to ensure that a magnetization held principally by hematite is purified of secondary overprints.

To ensure that we are able to recognize an "unshallowed" VGP distribution, we conduct this analysis on a sequence of published paleomagnetic results from Oligocene volcanic rocks from Yemen (i.e. from similarly low latitudes), as volcanic rocks are not affected by the inclination shallowing process (Fig. 7d; data from Riisager et al., 2005). Finally, to consider the possible bias introduced by small data-sets and/or the centering process, we: (1) collect a series of small data-sets ($N = 20$), randomly sampled from a larger population ($N = 500$) generated by TK03.GAD, and (2) calculate the elongation of combined groups of these data-sets (15 sets of $N = 20$) after centering them individually according to the procedure described above. We compare the

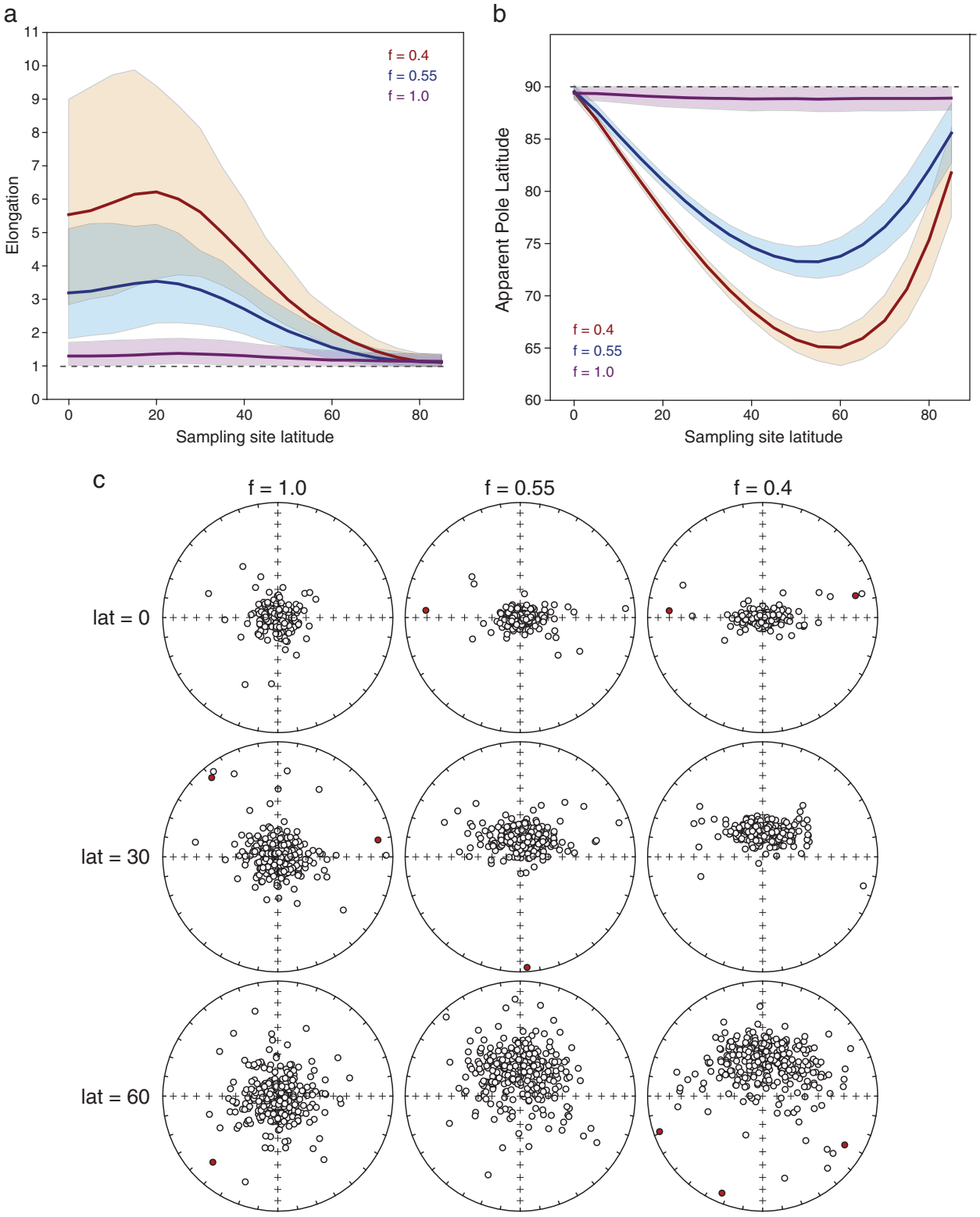


Fig. 5. Results of variable inclination shallowing on statistical field model-generated VGP distributions and their apparent pole latitudes. a) Elongation of VGP distribution vs. sampling site latitude, according to various shallowing factors: $f = 1.0$ (purple), 0.55 (blue), 0.4 (red). Elongation is defined as the ratio of the intermediate and minimum eigenvalues (τ_2/τ_3) of the orientation matrix of the VGPs; a circular distribution has an elongation of 1 (dashed line). The shallowing factor (f) relates an observed inclination (I_o) to the true inclination (I_f), according to: $\tan(I_o) = f\tan(I_f)$; $f = 1.0$ results in no shallowing. The solid curves are mean results, bounded by the colored areas which enclose 95% of the individual estimates. The results were determined by generating 300 directions from statistical field model TK03.GAD at each 5° interval of latitude, shallowing them by f, and measuring the elongation of the resulting VGP distribution. The mean estimates and 95% envelopes were determined after 1000 iterations of this process. b) Apparent pole latitude vs. sampling site latitude, after application of the various shallowing factors from a). The solid curves are mean results, bounded by the colored areas which enclose 95% of the individual estimates. The results were determined by the same process in a), but rather measuring the latitude of the resulting mean VGP in each individual iteration. Estimates and 95% envelope were determined after the process was repeated 1000 times. c) Example VGP distributions from TK03.GAD according to changes in sampling site latitude and f. Open (closed) symbols are projections onto the upper (lower) hemisphere. Note the increasing elongation and changing center of mass of the distributions as a function of f.

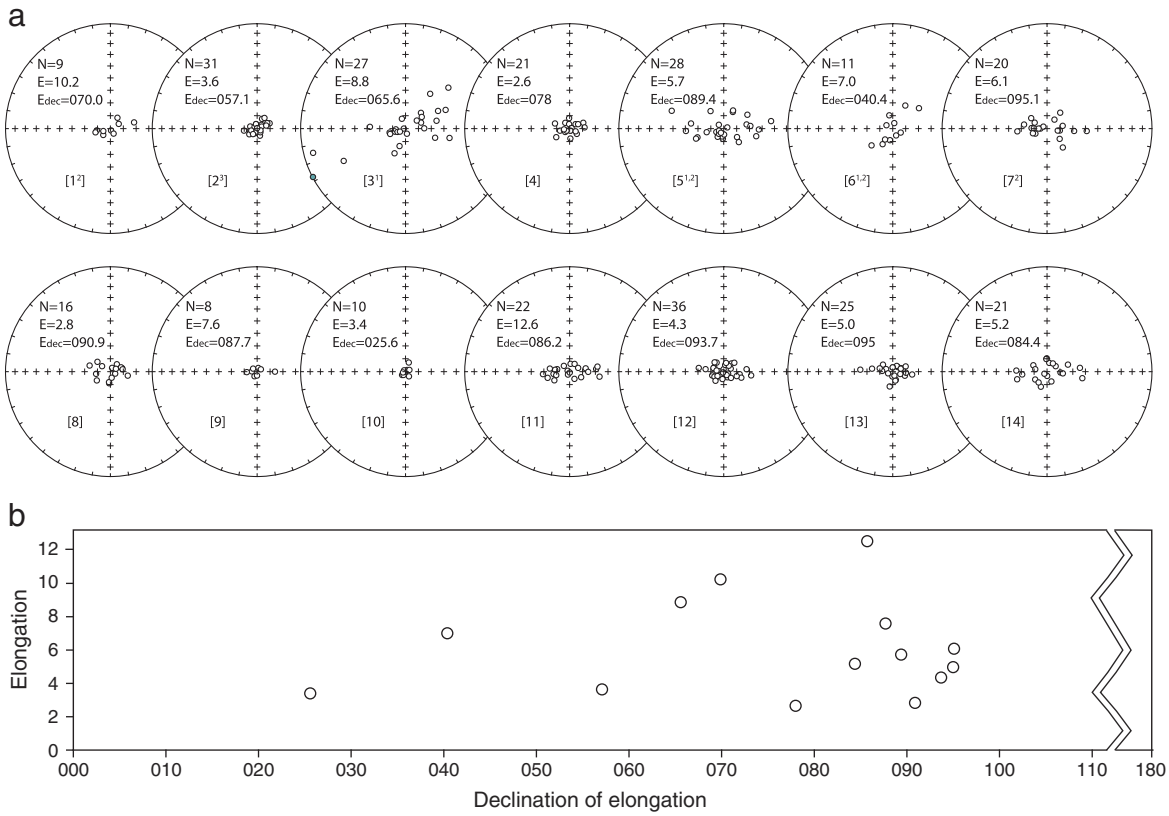


Fig. 6. VGP distributions of selected Laurentian paleomagnetic records (N≥8). a) VGP distributions after centering according to the procedure described in the text. All symbols are projections onto the upper hemisphere. N = number of site-level VGPs, E = elongation, E_{dec} = declination of elongation, [#] = reference number of record, 1: Wolfcampian units, Peterson and Nairn (1971), 2: Cutler Fm., Gose and Helsley (1972), 3: Lykins and Fountain Fms., McMahon and Strangway (1968), 4: Pictou redbeds, Symons (1990), 5: Lower Maroon Fm., McMahon and Strangway (1968), 6: Upper Maroon Fm., McMahon and Strangway (1968), 7: Dewey Lake Fm., Molina Garza et al. (2000), 8: Dewey Lake Fm., Molina-Garza et al. (1989), 9: Chugwater Fm., Herrero-Bervera and Helsley (1983), 10: Chugwater Fm., Shive et al. (1984), 11: Moenkopi Fm., Steiner and Lucas (1992), 12: Moenkopi Fm., Molina-Garza et al. (1991), 13: Moenkopi Fm., Molina-Garza et al. (1996), 14: Lower Fundy Group, Symons et al. (1989). [1]Results not adequately demagnetized (see text for explanation). [2]Results not included in the compilation of Torsvik et al. (2008). [3]ChRM explicitly interpreted as CRM by original authors (Gose and Helsley, 1972). b) Elongation of VGP distributions vs. declination of elongation for studies in a). Elongation estimates are likely unreliable due to the small size of the data-sets (N≤36).

elongation determined from several iterations of this experiment against the "true" elongation of the original population (Fig. 7e); the experiment was run both with $f=1$ and $f=0.4$.

7. Discussion

7.1. Interpretation of paleomagnetic results

We interpret the ChRM of the intrusive rocks and contact metamorphosed limestone to be a primary, Middle Permian (~270 Ma) magnetization. The exclusively reversed polarity of this remanence is consistent with magnetization acquisition during the Kiaman Reversed Superchron, which extended from ~318–265 Ma (Gradstein et al., 2004; Opdyke et al., 2000). The Middle Permian K–Ar and ^{40}Ar–^{39}Ar age determinations on the Grants Intrusive and Downeys Bluff Sill discount the possibility of a significant (>300°C) thermal event after ~270 Ma, which rules out a secondary magnetization of thermal origin. A secondary chemical remanent magnetization (CRM) is similarly unlikely due to the identification of titanomagnetite as the principal remanence carrier. Reynolds et al. (1997) observed magnetite–ilmenite intergrowths and TiO_2-rich high-temperature alteration products - both indicative of an igneous origin - in Grants Intrusive and Downeys Bluff Sill. Furthermore, we note that the position of the paleomagnetic pole resembles other Middle Permian poles from Laurentia, but not younger ones (Kent and Irving, 2010). We interpret the low-temperature/coercivity component (A) to be an overprint of recent origin. The paleomagnetic pole determined from this component (79.7°N, 349.7°E $A_{95}=18.4°$; in situ coordinates) resembles that of the present day field. The C-component observed in the Hart Creek Diatreme and interpreted to reside in hematite, is also interpreted as a recent overprint, perhaps developed by surface oxidation of primary titanomagnetite.

We observe a minor reduction in the estimated α_{95} of the mean paleomagnetic direction after tilt-correction (applied to data from the Chamberlain Diatreme and Grants Intrusive), but note that this is not statistically significant at the 95% confidence level, and does not constitute a positive tilt-test. Furthermore, the nature and timing of the tilting is not well-established. Host strata for the Willow Lake Dike and the Downeys Bluff Sill exhibit no significant structural dip (<5°). Strata near the Hart Creek Diatreme exposure exhibit a minor dip to the NW, but the cross-bedded nature of this poorly exposed sandstone precludes any confident structural restoration. Exposures of strata near Grants Intrusive and the Chamberlain Diatreme exhibit clear bedding contacts with a significant dip. The Chamberlain Diatreme occurs along the Herod Fault Zone, a NE-trending structure which was likely active after emplacement of the Diatreme, and may be responsible for this locally observed dip. Grants Intrusive is exposed on the southwest flank of Hicks Dome, which is assumed to have developed contemporaneously with the regional magmatism; thus, the structure may pre- or post-date the intrusion. We prefer the tilt-corrected result, due to the minor improvement in site-mean clustering.

7.2. Interpretation of VGP distribution analysis

It is evident from the example VGP distributions (Fig. 5c) that the shallowing of magnetic field directions results in an elongation of the

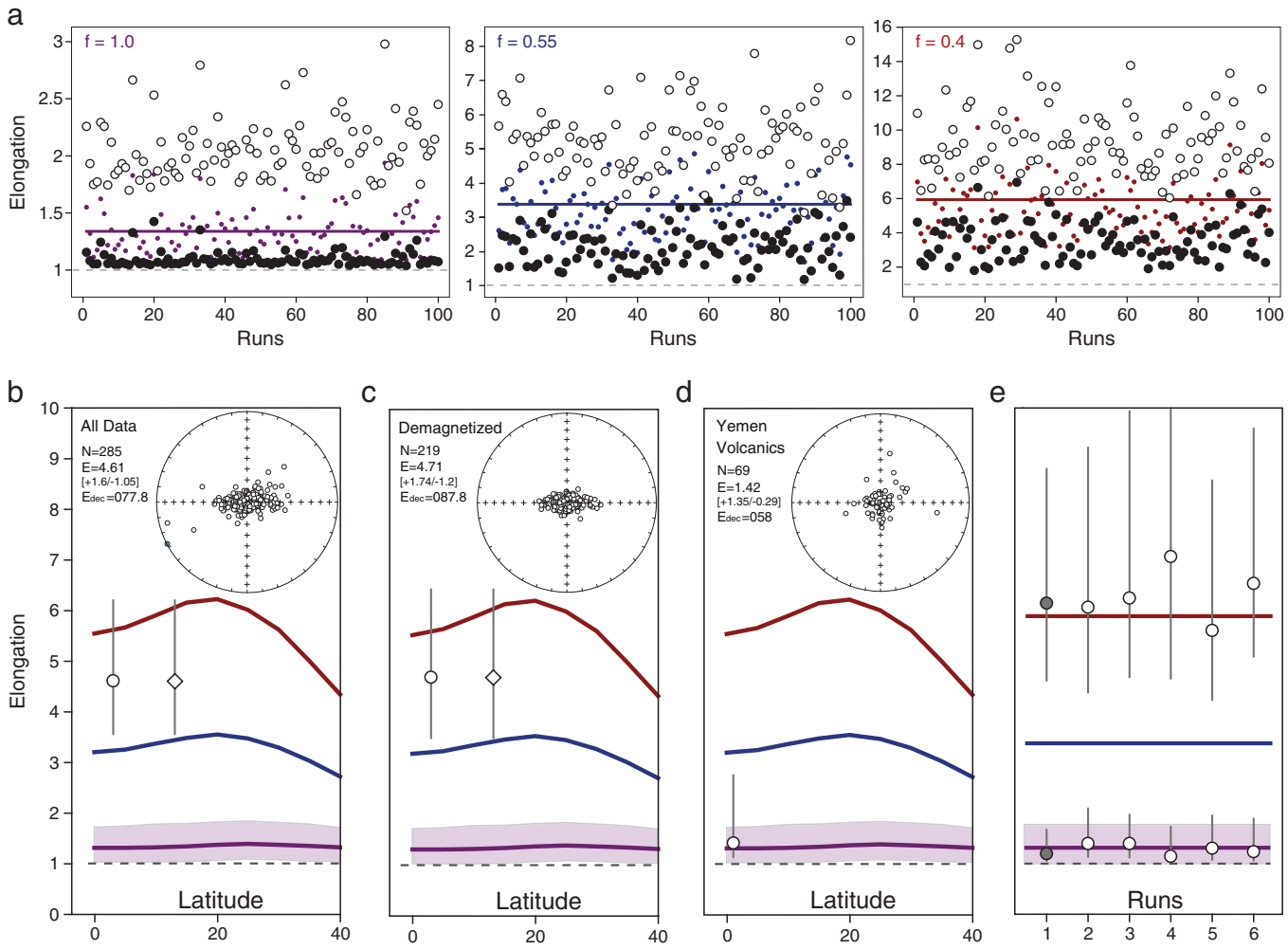


Fig. 7. Results of VGP distribution analysis. a) Example elongation estimates from 100 randomly generated data-sets (N = 300) at a latitude of 10°, under shallowing conditions of $f = 1.0$, 0.55, and 0.4 (note, the vertical scales are different). Small colored symbols are the mean elongation estimates, the larger solid (open) symbols are the minimum (maximum) elongation estimates from the 95% bootstrap confidence limits. The solid horizontal line is the mean elongation estimate determined from 1000 iterations (Fig. 5a); the dashed line represents a distribution with circular symmetry (elongation = 1.0). 95% of the individual estimates overlap with the mean estimate (within error). b–d) Elongation estimates of published VGP records. Circles (diamonds) depict mean elongation estimates vs. sampling site latitude determined from the uncorrected (max. corrected) mean inclination. The vertical gray lines show the 95% bootstrap confidence limits on the mean elongation estimates. The solid curves in each panel are taken from Fig. 5a (purple: $f = 1.0$, blue: $f = 0.55$, red: $f = 0.4$). The inset stereonets show the VGP distributions. N = number of VGPs, E = mean elongation (95% confidence bounds are in brackets), E_{dec} = declination of elongation. b) All data from Fig. 6a. c) Data from b) after removal of poorly demagnetized records (i.e. the records from McMahon and Strangway (1968); denoted by [1] in Fig. 6a). d) Results from Oligocene volcanic rocks from Yemen; data from Riisager et al. (2005). e) Test for bias introduced by small data-sets and/or the centering process described in the text. The gray circles are elongation estimates determined from a data-set with N = 500, generated by TK03.GAD at a latitude of 10° and a mean declination of 0°. The open circles are elongation estimates determined from a collection of small (N = 20), randomly sampled subsets of the initial data which are individually rotated according to the centering process before being grouped together (into15 sets of N = 20 for a grouped collection of N = 300) and collectively analyzed. The experiment was conducted using both $f = 1.0$ and $f = 0.4$. The consistency of the elongation estimates suggests that no significant bias is introduced by combining small data-sets or centering the VGP distributions.

VGP distribution along meridians perpendicular to those containing the sampling site. In the model, the site longitude was set to 0°, so the flattening of directions has expectedly resulted in an elongation of the VGP distribution along the 90°–270° meridians. Because all of the Laurentian data-sets have been rotated into the same reference frame (sampling site longitude = 0°), the incidence of inclination shallowing in these records should be reflected in an elongation of the VGP distribution in the same sense. Fig. 6 illustrates a clear 90° (=270°) preference in the elongation direction of these data-sets, in support of the hypothesis that inclination shallowing has affected the paleomagnetic records from these redbeds. The grouped results in Fig. 7b,c similarly exhibit a clear 90°–270° elongation of the VGP distribution. In both grouped analyses, the calculated value of elongation is statistically distinct (95% conf.) from a circular distribution (elongation = 1), indicating that significant shallowing of the inclinations has occurred. This conclusion is reinforced by the analysis of the Yemeni volcanic rocks, which yield a very low VGP elongation, in agreement with the expectation that unshallowed directions (from a GAD) should produce a circularly symmetric VGP distribution (Fig. 7d). Our exercises in sub-sampling and centering synthetic data have resulted in elongation estimates that are reasonably close to (and statistically indistinguishable from) the elongations derived from the original data and the model expectations, thereby implying that our sample sizes and centering procedure are not introducing any systematic bias (Fig. 7e).

Beck et al. (2003) conducted a similar VGP distribution analysis of paleomagnetic data from Triassic and Jurassic redbeds from the Colorado Plateau, but concluded that the observed elongations were due to a protracted magnetization acquisition. Cederquist et al. (1997), examining the elongation of Carboniferous–Early Jurassic magnetization directions from Laurentia, also concluded that the observed elongate distribution was due to the inclusion of apparent polar wander in the data as a consequence of protracted magnetization acquisition. This alternative interpretation is predicated on the assumption that the ChRM

of these redbeds is dominantly chemical in nature, and was acquired over a prolonged period by the slow growth of authigenic (pigmentary) hematite in the form of a grain-coating/cement. However, a review of the original studies shows that the contribution of pigmentary hematite to the ChRM is often minor (if existent), relative to the contribution by specular hematite, of detrital origin. Such a depositional remanent magnetization (DRM) is supported by the lateral continuity of polarity intervals across hundreds of kilometers and across lithologic boundaries (Purucker et al., 1980; Steiner et al., 1993), sequential changes in magnetization through thin sedimentary units (Shive et al., 1984), positive intraformational conglomerate and soft-sediment fold tests (Elston and Purucker, 1979; Magnus and Opdyke, 1991; Molina-Garza et al., 1991; Symons, 1990; Symons et al., 1989), co-linear magnetization directions with detrital magnetite (Steiner, 1988), characteristic specularite demagnetization behavior (Collinson, 1974; Gose and Helsley, 1972), and by petrographic observations (Elston and Purucker, 1979). More broadly, recent work on Carboniferous and Late Triassic–Early Jurassic redbeds from North America has demonstrated a common shallow inclination bias, implying that the ChRM of these redbeds is predominantly a DRM (Bilardello and Kodama, 2010; Kent and Irving, 2010; Kent and Tauxe, 2005). Following our review of the original studies, we have labeled the records that present evidence consistent with a DRM as the ChRM (Table 2). We have also labeled the results that the primary authors explicitly associated with a CRM, although we stress that these determinations are not conclusive. If most of the paleomagnetic records are derived from DRMs, as we argue the evidence suggests, then the agreement between VGP distribution elongation and Laurentian APW, as observed by Cederquist et al. (1997) and Beck et al. (2003), is merely coincidental. Indeed, the late Paleozoic–early Mesozoic APWP of Laurentia is roughly orthogonal to its central meridian, so that most sampling sites with a shallow inclination bias could be expected to yield a VGP distribution with an elongation apparently coincident with the APWP. We consider most of the redbed paleomagnetic records to be depositional in nature, and interpret the elongate VGP distributions to be a consequence of inclination shallowing.

7.3. Magnitude of shallow inclination bias

The obvious reduction in VGP scatter between panels b and c in Fig. 7 is an indication of the presence of unremoved magnetization overprints in the data-sets of McMahon and Strangway (1968). As such, we regard the result in panel c to be the most representative of the primary records, although the change in the elongation estimate between the panels is negligible. The elongation determined from the data in panel c is 4.71; the bootstrap confidence limits extend from 3.51 to 6.45. Numerical exercises conducted on the VGP sets generated by TK03.GAD have demonstrated that the 95% bootstrap confidence limits on the estimated elongation of any given data-set have a 95% probability of overlapping with the "true" mean VGP elongation (from Fig. 5a), provided that the data-set is sufficiently large ($N \geq 100$–150) (Fig. 7a). This is an expected outcome where both the "true" mean VGP elongation is well-determined and the individual sample-sets fully characterize the short-term directional variance in the field (model). Because our combined Laurentian data-set is sufficiently large, we may exclude all theoretical shallowing factors (f) that do not produce mean VGP distribution elongations that overlap with the bootstrap confidence bounds on the elongation estimate from the Laurentian data. If we use the *uncorrected* mean inclination of the data-set to calculate the sampling site latitude, our elongation estimate (with error) suggests that shallowing by a factor of $f=0.52$ to $f=0.37$ has affected the data-set (circle, Fig. 7c). However, this approach is not strictly correct, as the theoretical curves are estimates of VGP distribution elongation for various f values at the true sampling site latitude (i.e., determined from the *corrected* mean inclination, which is typically unknown). Instead, a recursive correction–calculation process can be employed, wherein the maximum and minimum allowable inclination corrections can be determined according to those f values that would still lie within the confidence bounds of the elongation estimate after the inclination correction is performed. The highest sampling site latitude allowed by the data (by the maximum inclination correction consistent with the elongation estimate) is depicted by a diamond in Fig. 7b,c. The shallow nature of the theoretical curves at low-latitudes and the low-mean inclination (5.3°) of the grouped data-sets precludes any dramatic shift in the range of potential f values; applying the recursive inclination corrections changes the potential range to: $f=0.54$ to $f=0.39$, with a mean of $f=0.46$. For reference, Bilardello and Kodama (2010) compiled published f values determined from hematite-bearing sedimentary rocks. They range from $f=0.83$ to $f=0.4$, with a mean value of $f=0.59$. More specifically, Kent and Tauxe (2005) determined eight f values from Late Triassic redbeds from North America, which ranged from $f=0.66$ to $f=0.4$, with a mean value of $f=0.56$.

7.4. Implications for APWP of Laurentia

Although the IL pole resembles other Middle Permian poles from Laurentia, it does exhibit a significantly higher paleolatitude, as evident by its position relative to the Middle Permian segment of the Laurentian APWP of Torsvik et al. (2008) (Fig. 8a). Due to the prevalence of paleomagnetic records from continental redbeds in the compilation by Torsvik et al. (2008), we argue that this discrepancy is due to a widespread shallow inclination bias in the poles derived from sedimentary rocks. Our VGP distribution analysis supports this argument, in that the compiled data exhibit an elongation that is statistically distinct (95% conf.) from the expectations of a VGP distribution generated by a GAD. Our conservative estimate of elongation suggests that an average shallow inclination bias equivalent to shallowing by $f=0.54$ is present in the paleomagnetic records of these continental redbeds.

To consider the effect that an inclination correction will have on the Laurentian APWP, we have first updated the 300–235 Ma compilation of Torsvik et al. (2008), according to the 2009 geologic timescale (Table 2). Of the five igneous results that were reported in the original compilation, only one result (#37) remains after updating the ages; and this result is derived from a single intrusion (i.e. single cooling unit), highlighting the importance of the new IL result. We have also added an Early Triassic result from a diatreme in British Columbia (#36), but this result is also based on a single cooling unit. In Fig. 8b we plot the updated APWP after correcting the inclination of all redbed records with an f value of 0.54. There is an obvious improvement in the agreement between the IL result and the Middle Permian segment of the APWP after these corrections. In Fig. 8c we correct the redbed records according to $f=0.46$, which corresponds to the mean estimated shallowing factor determined from our VGP distribution analysis. This correction does not appreciably improve the agreement between the IL pole and the APWP relative to the more conservative correction of $f=0.54$. We note the increasing A_{95} of the mean poles with increasing inclination correction (decreasing f), and attribute this to our simplistic "average" correction. Although we have argued that the redbeds are all likely affected by a shallow inclination bias, it is unlikely that they all share a bias of exactly the same magnitude. We contend that our analysis is a valuable demonstration of the widespread presence of too-shallow inclinations in these records, and that our correction offers a better approximation of the true Laurentian APWP, but future detailed work will be necessary to more precisely correct the results for each record. Due to the correlation between mean pole A_{95} and f and the lack of significant improvement between the corrections of $f=0.54$ and $f=0.46$, we prefer the conservative correction.

Regardless of the improvement between the IL pole and the Laurentian APWP, the shift in the APWP after inclination correction brings it into better agreement with the APWP of Gondwana. After updating the ages of the poles used in the Laurentian compilation, the APWP also exhibits a sharp bend in the latest Paleozoic, as previously observed in Gondwana's APWP (Domeier et al., 2011). The discrepancy remaining between the APWPs in Fig. 8b may be ascribed to a similar shallow

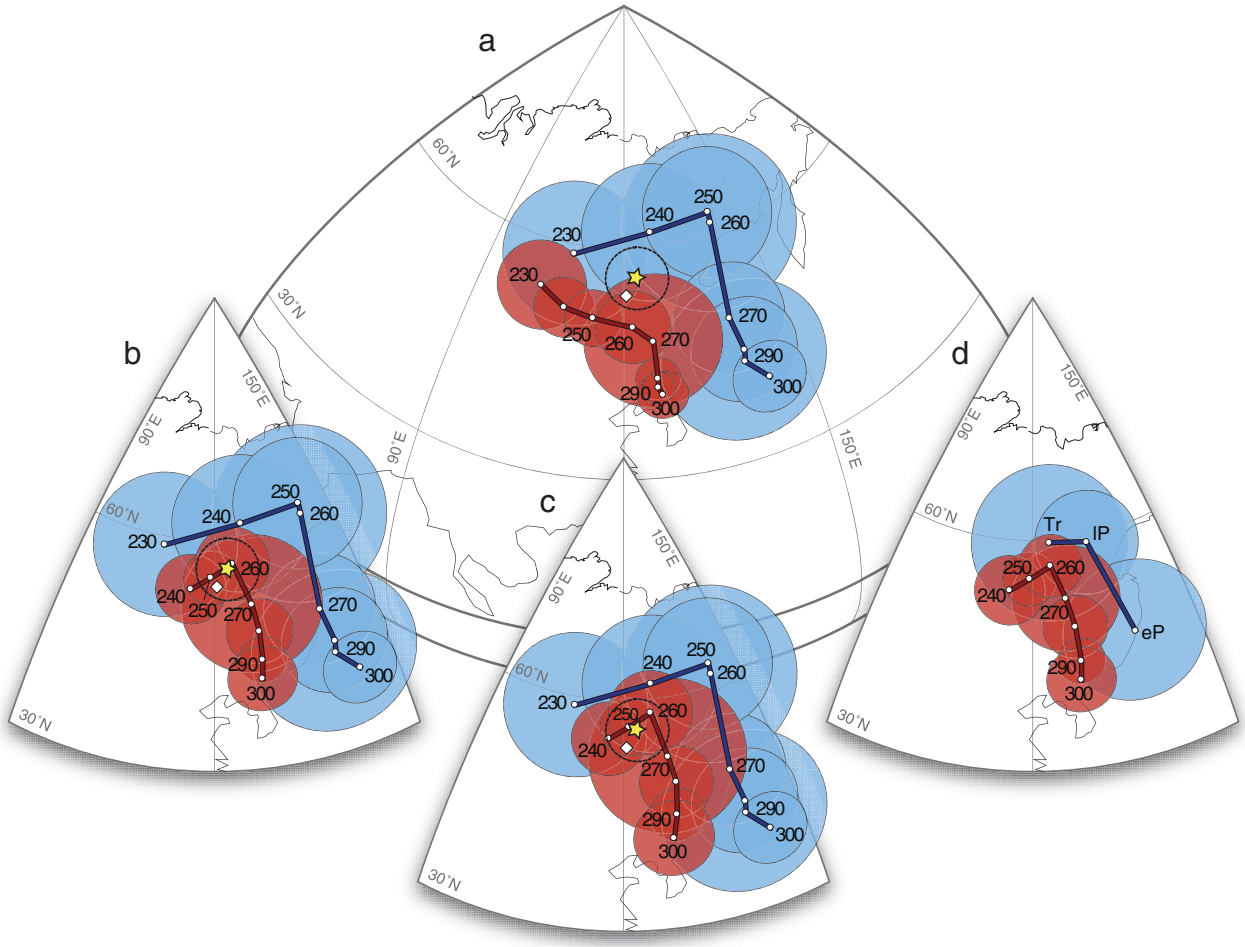


Fig. 8. Effect of inclination shallowing corrections on the Laurentian APWP. a) APWPs of Laurentia (red) and Gondwana (blue) plotted in North American coordinates after the paleomagnetic compilation and Pangea reconstruction parameters of Torsvik et al. (2008). The star (diamond) depicts the tilt-corrected (in situ) IL pole position, the dashed line is the A_{95} of this result. Note the large discrepancy between the Middle Permian to Early Triassic mean poles of the APWPs, and the location of the IL result relative to the APWPs. b) As in a), but after updating the ages of the Laurentian paleomagnetic data and applying an inclination correction using $f = 0.54$ (conservative estimate from Fig. 7c). Note the improvement between the Middle Permian segment of the APWP and the IL result. c) As in b), but after application of an inclination correction using $f = 0.46$ (mean estimate from Fig. 7c). d) Laurentian APWP from b) (corrected by $f = 0.54$) compared with an APWP constructed from a filtered paleomagnetic data-set from South America (Domeier et al., 2011), rotated into North American coordinates according to the reconstruction parameters of Torsvik et al. (2008).

inclination bias — among other data-pathologies — in the paleomagnetic records from Gondwana. In Fig. 8d we compare the f = 0.54 inclination corrected APWP of Laurentia with a filtered Permian and Triassic paleomagnetic data-set from South America (Domeier et al., 2011). Although this filtered data-set is not explicitly corrected for inclination shallowing, it includes more poles based on data from igneous rocks than the larger Torsvik et al. (2008) compilation, and the included records have been subjected to more stringent selection criteria. The separation between the Late Permian pole from the filtered South American data-set and the Late Permian poles of the inclination corrected APWP of Laurentia in Fig. 8d is greatly reduced relative to Fig. 8a, supporting the contention that the long-standing discrepancy between the APWPs of Laurussia and Gondwana may simply be a manifestation of systemic bias.

8. Conclusions

Our investigation of shallow intrusive rocks and a contact metamorphosed limestone in southern Illinois (cratonic Laurentia) yields a new Middle Permian (~270 Ma) paleomagnetic pole with a paleolatitude that is distinctly higher than that of the Middle Permian segment of the reference APWP of Laurentia. We argue that this discrepancy is due to a widespread, shallow inclination bias in the Laurentian paleomagnetic data, following the observation that the reference APWP is dominated by data derived from continental redbeds. A comparison between theoretical VGP distributions and published VGP sets from the Laurentian redbeds reveals the latter to have a distribution that is incompatible with the expectations of TK03.GAD. The commonly elongate VGP distributions observed in the Laurentian redbed paleomagnetic records are instead consistent with a pervasive shallow inclination bias, suggesting that a "flattening" on the order of $f = 0.54$ to $f = 0.39$ (mean of $f = 0.46$) has affected the inclinations. Using the conservative estimate of shallowing ($f = 0.54$), we correct the inclinations of the redbed paleomagnetic records and demonstrate that the modified Laurentian APWP agrees more closely with both our new igneous rock-based paleomagnetic pole and the APWP of Gondwana. We contend that such a shallow inclination bias is also likely to be widespread in the sedimentary rock-based paleomagnetic records from Gondwana, and that this data-pathology may be the principal cause of the long-standing disparity between the APWPs. Finally, we note that the approach adopted here is simplistic in its assumption that all the redbeds have experienced the same degree of shallowing; future detailed work on individual units will be necessary to refine our estimates.

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